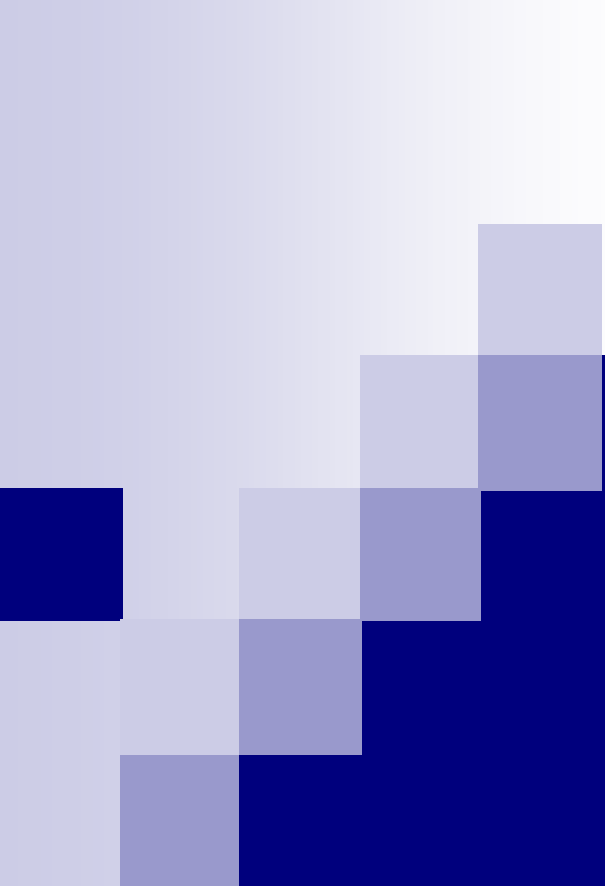




# Software Quality Assurance

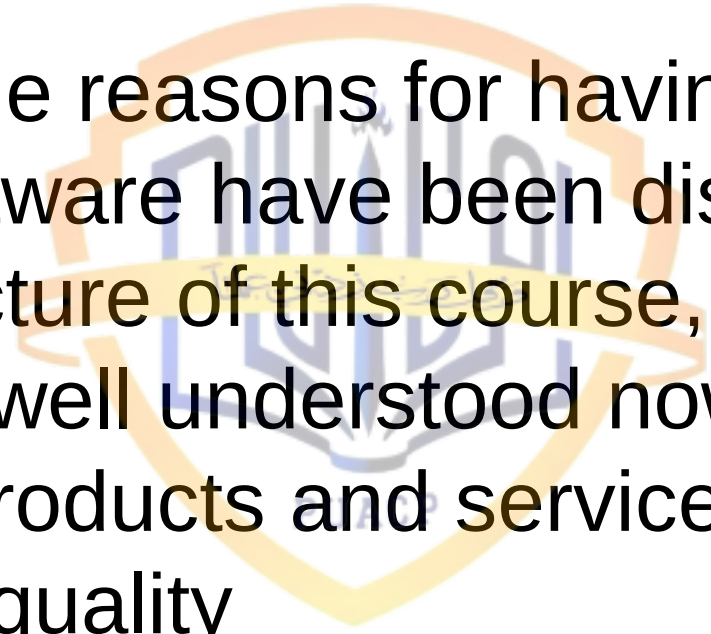


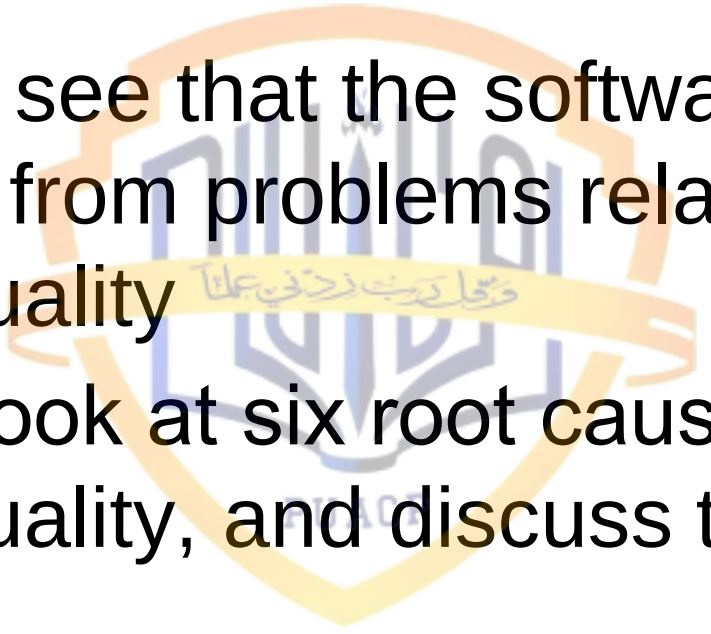
## Lecture # 4

# Today's Lecture

- Reasons for poor software quality
- Characteristics of quality laggard companies
- Quality attributes



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- Some of the reasons for having high quality software have been discussed in the first lecture of this course, and so it should be well understood now why software products and services should have high quality
  - There are negative consequences for poor or bad quality software

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- But still we see that the software industry still suffers from problems related to software quality
  - Now we'll look at six root causes of poor software quality, and discuss them in detail

# Root Causes of Poor Software Quality - 1

- Inadequate training of managers and staff
- Inadequate defect and cost measurement
- Excessive schedule pressure

# Root Causes of Poor Software Quality - 2

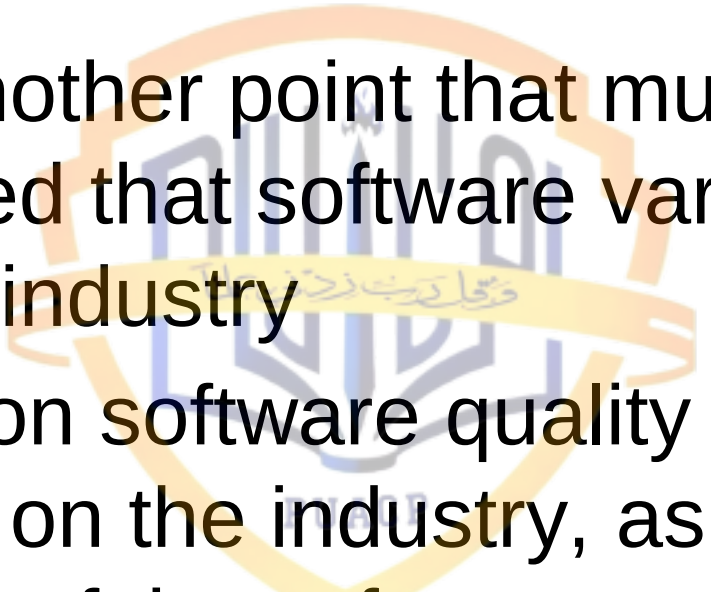
- Insufficient defect removal
- High complexity levels
- Ambiguous and creeping requirements and design (*feature race & gimmicks*)

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- We have just finished the discussion on root causes of poor software quality
  - Now let us look at the status of the software industry's seriousness of the software industry with respect to the software quality assurance

# Quality Assurance Organizations

- No quality assurance 60%
- Token quality assurance 20%
- Passive quality assurance 15%
- Active quality assurance 5%



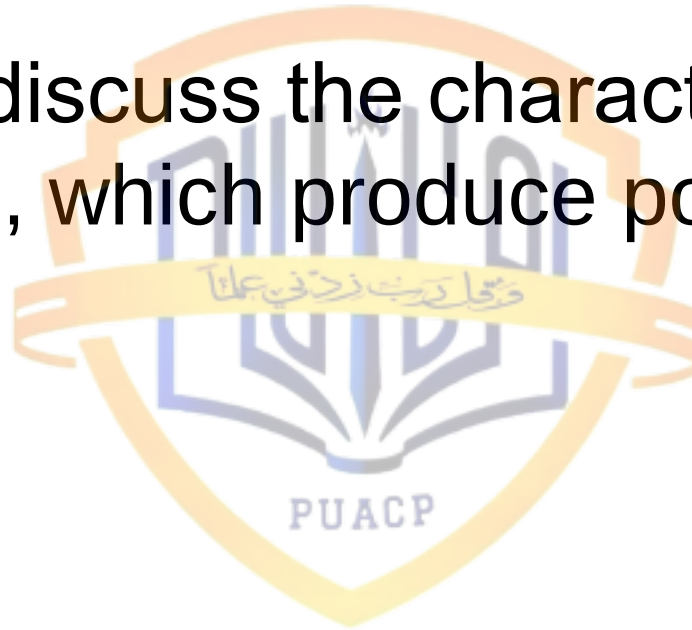
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- There is another point that must be remembered that software varies from industry to industry
  - The focus on software quality naturally is dependent on the industry, as well as the importance of the software application. More critical applications, naturally, need to have higher software quality than others

# Software Quality in Six Sub- Industries

- Systems software that controls physical devices
- Information systems that companies build for their own use
- Outsource or contract software built for clients
- Commercial software built by vendors for lease or sale
- Military software built following various military standards
- End-user software built for private use by computer literate workers or managers

# Characteristics of Quality Laggards

- We'll now discuss the characteristics of companies, which produce poor quality software



# Quality Laggards - 1

- No software quality measurement program of any kind
- No usage of formal design and code inspections
- No knowledge of the concepts of defect potentials and defect removal efficiency

# Quality Laggards - 2

- Either no quality assurance group or a group that is severely understaffed
- No trained testing specialists available
- Few or no automated quality assurance tools
- No quality and reliability estimation capability

# Quality Laggards - 3

- Minimal or no software project management tools available
- No automated risk assessment or avoidance capability
- From a low of one to a high of perhaps four distinct testing stages

# Quality Laggards - 4

- No test library or test-case management tools available
- No complexity analysis tools utilized
- Defect potentials averaging more than 6 defects per function point

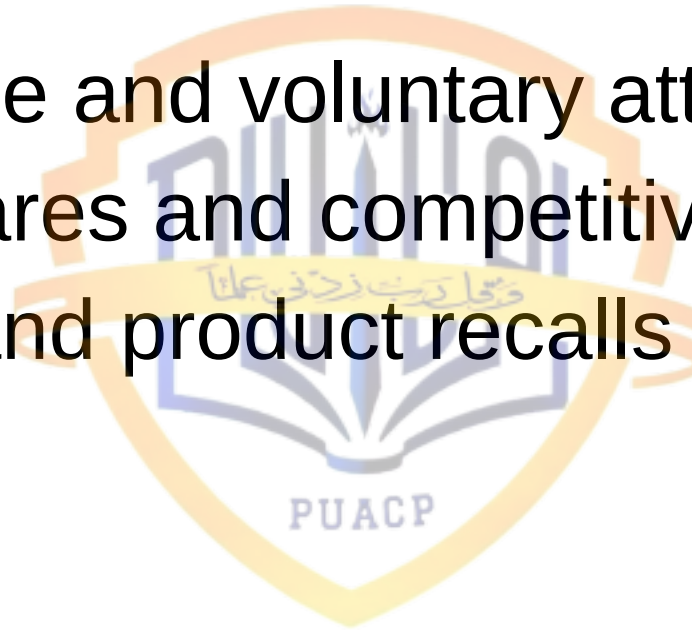
# Quality Laggards - 5

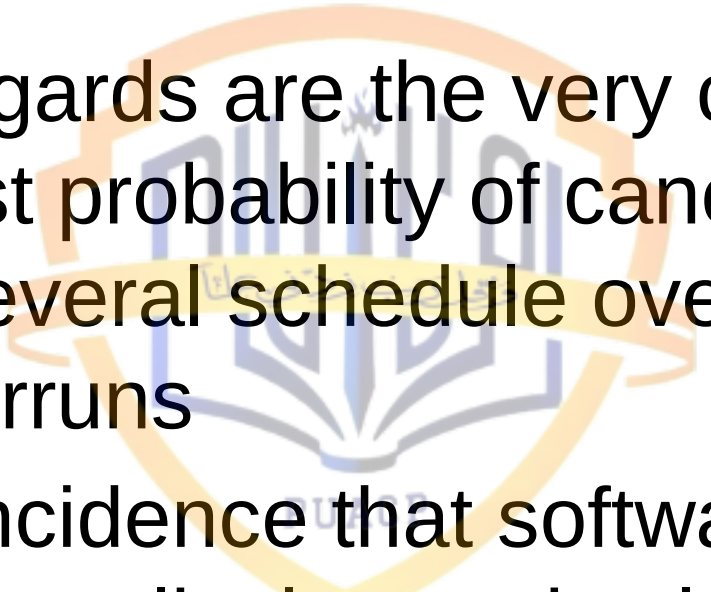
- Defect removal efficiency averaging less than 80%
- Executive and managerial indifference (and ignorance) of quality matters

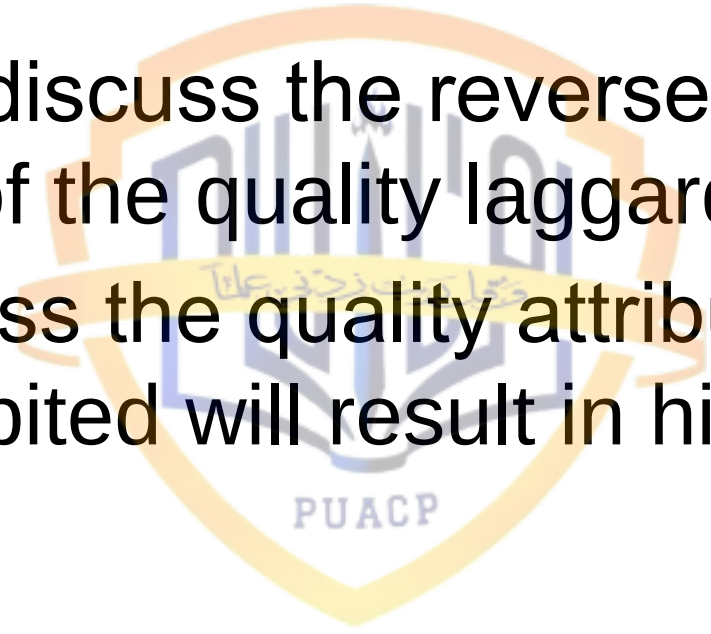


# Other Related Issues

- Staff morale and voluntary attrition
- Market shares and competitive positioning
- Litigation and product recalls



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- Quality laggards are the very companies with highest probability of cancelled projects, several schedule overruns, and severe overruns
  - It is no coincidence that software groups among the quality laggards also tend to be candidates for immediate replacement by outsource organizations

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- Let's now discuss the reverse of the practices of the quality laggards
  - Let's discuss the quality attributes, which when exhibited will result in high quality software

# Quality Attributes

- Quality attributes set is a way to represent customer quality requirements
- Ask your current and prospective customers about their definition of quality
- Develop a quality assurance program based on the requirements of your customers

# Categories of Quality Attributes

- Product-specific quality attributes
- Organization-specific quality attributes



# Product-Specific Attributes - 1

- Ease of use
- Documentation
- Defect tolerance
- Defect frequency



# Product-Specific Attributes - 2

- Defect impact
- Packaging
- Price versus reliability
- Performance



# Organization-Specific Attributes

- Service and support
- Internal processes





# Achieving High Levels of Software Quality - 1

- Enterprise-wide quality programs
- Quality awareness and training methods
- Quality standards and guidelines
- Quality analysis methods
- Quality measurement methods

# Achieving High Levels of Software Quality - 2

- Defect prevention methods
- Non-test defect removal methods
- Testing methods
- User-satisfaction methods
- Post-release quality control

# Best in Class Quality Results - 1

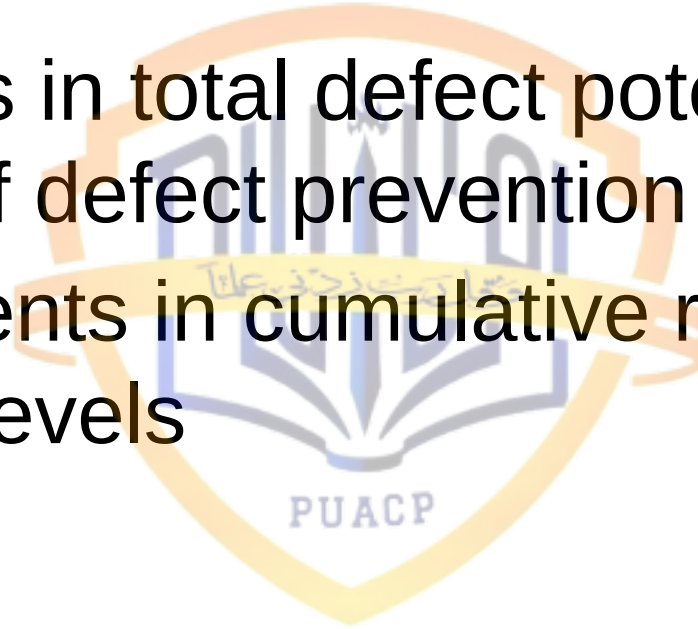
- Quality measurements
- Defect prevention
- Defect and quality estimation automation
- Defect tracking automation
- Complexity analysis tools

# Best in Class Quality Results - 2

- Test coverage analysis tools
- Formal inspections
- Formal testing by test specialists
- Formal quality assurance group
- Executive and managerial understanding of quality

# Two Components of Software Quality Improvement

- Reductions in total defect potentials using methods of defect prevention
- Improvements in cumulative removal efficiency levels



# References

- Software Quality: Analysis and Guidelines for Success by Capers Jones
- Customer-Oriented Software Quality Assurance by Frank Ginac